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Isolation and identification of quorum sensing antagonist from *Cinnamomum verum* leaves against *Pseudomonas aeruginosa*

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A B S T R A C T

Purpose: The study aimed at isolating and identifying potential anti-quorum sensing (QS) compounds from *Cinnamomum verum* leaves against *Pseudomonas aeruginosa*.
Methodology: Isolation of anti-QS compounds from *C. verum* leaf ethanol extract was carried out by column chromatography. The bioactive fraction was analysed by UV, IR, and GCMS spectroscopy. Various virulence assays were performed to assess the QS quenching ability of the purified compounds. *In vivo* toxicity of the purified compounds was examined in zebrafish model. The expression of the virulence genes was evaluated by qPCR analysis and *in silico* assessment was accomplished to check the binding ability of the compounds with the autoinducer molecule.
Key findings: The QS inhibitors isolated and identified showed a remarkable ability in reducing the production of elastase, pyocyanin, swarming motility and biofilm formation in *P. aeruginosa*. In the presence of the characterized compounds, the expression of virulence genes of *P. aeruginosa* was significantly reduced. Toxicity studies in zebrafish model indicated no effects on development and organogenesis at a concentration below 100 mg/l. Further, *in silico* analysis demonstrated the binding efficiency of the anti-QS compounds to AHL molecules, thus proving the QS quenching ability of the isolated compounds.
Significance: To the best of our knowledge this is the first report of isolation of anti-QS compounds from *C. verum* leaves against *P. aeruginosa*. The identified compounds qualify as potential QS antagonists. Further studies on these compounds can pave way for an effective and attractive anti-pathogenic therapy, to overcome the emergence of antibiotic resistance in bacteria.

1. Introduction

Over the last few decades, multidrug-resistant bacterial infections have become one of the major concerns globally resulting in increased mortality, cost of treatment and prolonged hospital stay. *Pseudomonas aeruginosa* is one such multidrug-resistant opportunistic nosocomial pathogen that causes life-threatening infections among the debilitated individuals [1]. Centres for disease control and prevention in their 2019 report has described multidrug-resistant *P. aeruginosa* as a serious threat. The severity of *P. aeruginosa* infection is due to the production and secretion of various virulence factors such as elastase, pyocyanin, exotoxins, alkaline protease, rhamnolipids and siderophores together with its ability to form biofilms on biotic and abiotic surfaces. The production of these virulence factors in *P. aeruginosa* is under the control of quorum sensing (QS) system.

QS refers to the ability of bacteria to converse among themselves

Abbreviations: algD, alginate; *C. violaceum*, *Chromobacterium violaceum*; *FlagA*, flagellar basal body protein A; FTIR, Fourier-transform infrared spectroscopy; GCMS, gas chromatography mass spectrometry; hpf, hours post fertilization; *LasB*, elastase B; *P. aeruginosa*, *Pseudomonas aeruginosa*; *PhzH*, phenanzine; *PiliA*, Type IV Pili; *rhlI*, acyl homoserine lactone synthase; TLC, thin layer chromatography.

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when present in a group and is attained by small self-generated signalling molecules produced by the bacteria known as autoinducers. The production of these auto inducers is directly proportional to the bacterial cell density [2]. The autoinducers produced in *P. aeruginosa* are acylhomoserine lactones (AHLs), which are regulated by *LuxR* and *LuxI* genes. The *luxI* genes encode the enzyme required for the synthesis of AHL, and the cognate *luxR* gene encodes an intracellular receptor specific for AHL. Upon binding to the AHL signal, the receptor regulates the transcription of target genes [3]. Since QS is involved in controlling the production of several virulence factors and processes, attenuation of the QS system would be an attractive target to combat antibiotic resistance. Discovery of anti-QS agents, which could inhibit or disrupt AHL signals to attenuate QS system, is the need of the hour. Even though many anti-QS agents have been synthesized, none of them have been commercialised due to their reactivity and toxicity against eukaryotic cells or due to their reduced shelf life [4].

Plants represent the richest available reservoir of new therapeutic products. Since the 1990s, there has been a gradual shift in interest towards plant-based compounds as sources for new pharmaceuticals because of their alluring repertoire of phytochemicals. But reports on isolation of anti- QS compounds from locally available vegetables and spices against *P. aeruginosa* are limited. This prompted us to take up the research work to screen different vegetables and spices for their anti-QS activity. Of the 65 different vegetables and spices used, *Cinnamomum verum* crude leaf extract exhibited the highest inhibition of virulence factor production in *P. aeruginosa* [5]. This provided the basis for the selection of *Cinnamomum verum* leaves for the isolation and characterization of anti-QS compounds.

2. Methods

2.1. Bacterial strains

Pseudomonas aeruginosa, ATCC 15692 (from infected wound), a reference strain resistant to all β lactam drugs and *Chromobacterium violaceum* MTCC2656, a biomonitor strain were used in this study.

2.2. Plant material and preparation of extracts

Leaves of *Cinnamomum verum* J. Presl were used and the extraction was carried out according to the method described in our earlier study [5].

2.3. Fractionation of ethanol extract of C. verum

Anti-QS compounds found in the ethanol extract were subjected to solvent separation with solvents of increasing polarity *viz.*, petroleum ether, chloroform, and water. One gram of crude ethanol extract was mixed with 1ml of ethanol on the day of the experiment. This mixture was taken in a separating funnel with 100 ml of petroleum ether and 100 ml of distilled water. The mixture was shaken well and allowed to stand for 1530 min. The lower aqueous layer and upper petroleum ether layer were collected separately. The aqueous layer was once again mixed with 100 ml chloroform, shaken well and allowed to stand. The lower chloroform and the upper aqueous layers were collected. The separated fractions of petroleum ether, chloroform and water were flash evaporated to remove the solvent. Once dry, they were checked for their anti-QS activity against *C. violaceum* by agar well diffusion method. The procedure was repeated to collect a sufficient quantity of the extract for further experiments.

2.4. Column chromatography

Chromatography was performed according to the method of Policegoudra et al. [6] with minor modifications. Activated silica gel (60–120 mesh) with petroleum ether was loaded into a glass column (600 mm). Five grams of the fractionated dried extract with anti-QS activity was triturated with 1 g of silica and filled on top of the column packed with silica gel. Column elution was performed with petroleum ether, chloroform, ethanol and water with a flow rate of 1 ml/min. First elution was performed with a mixture of petroleum ether and chloroform mixed in different ratios such as 9:1, 8:2, 7:3, 6:4, 5:5, 4:6, 3:7, 2:8, 1:9 followed by chloroform: ethanol and ethanol: water mixed in the same ratios as mentioned above.

2.5. Thin layer chromatography

TLC strips (silica gel on aluminium foils L * W 20 cm * 20 cm, Sigma) were used. Petroleum ether and chloroform in the ratio of 6:4 were used as the mobile phase. Retention factor (Rf) value for each 5th tube elute of the fractionated extract was measured and visualized in the iodine chamber. Elute with the same Rf values were mixed and assessed for their anti-QS activity against *C. violaceum*.

2.6. Lambda max

One mg of the fractionated extract dissolved in 3 ml ethanol was placed in the sample chamber of the Spectroquant® Prove600 UV/VIS spectrophotometer (Merck, USA). The spectra were recorded in the range of 200–800 nm. Important absorption frequencies that appeared in the spectra were noted down.

2.7. Infra-red spectroscopy

Dried powder (0.5 mg) of the fractionated extract was placed in the sample chamber of the Alpha FT-IR spectrometer (Bruker, USA) and the spectra were recorded in the range of 4000–600 cm^{-1}. Important absorption frequencies that appeared in the functional group region of the spectra were recorded.

2.8. Gas chromatography mass spectrometry (GCMS)

Bioactive fractions were analysed in the Shimadzu GC–MS-138 QP2010S, containing Rxi®-5ms gas chromatography column (ID 0.25 mm, 0.25 μm thickness), coupled to a mass spectrometer mass detector (with 5% diphenyl and 95% dimethyl polysiloxane). The oven temperature was set at 60 °C with 2 min hold and was increased to 310 °C at the rate of 10 °C/min with 13 min hold. The injector temperature was set at 250 °C. The ionization voltage was 70 eV and helium was used as the carrier gas with a flow rate of 1 ml/min. The injection volume was 1 μl and the scan range was 45−450 amu. The total run time was 40 min. The major constituents of the bioactive fractions were identified by comparing with the MS reference database of NIST software (National Institute of Standards and Technology, Gaithersburg, USA).

2.9. Assessment of virulence of P. aeruginosa in the presence of purified compounds

Inhibition of swarming motility, biofilm formation, pyocyanin, and lasB elastolytic activity was checked in the presence of the purified compounds as previously described [5].

2.10. Toxicity studies of the purified compounds

The toxicity of the purified compounds was checked in the zebrafish model system. 250 mg of the purified compounds dissolved in 10 ml of dimethyl sulfoxide was used as the test sample for the experiment. The LC_{50} of the purified compounds was determined according to the method of Karber [7]. Different concentrations of the compound, ranging from 50 mg/l to 500 mg/l were used for toxicity studies. A semi-static methodology, where media change with renewal of test compound twice a day, was followed. Embryos were maintained at 28 °C in E3 medium. The embryos were exposed at 10 hours post fertilization (hpf) and were monitored for five days for developmental and organ-specific toxicity [8]. The assessment at different stages of development was done for a total of 22 toxicological lethal/sublethal endpoints (Table S1). Each endpoint was scored in a binary fashion (present/absent) and was recorded as percent incidence relative to total embryos at a given concentration. Data from three replicates (20 embryos in each replicate) were recorded for each endpoint and then compared with the control (unexposed) embryos. The morphological examinations were done using a stereo-zoom trinocular microscope (S9D, Leica). The embryos exposed with equivalent volume of DMSO without the test compound served as control.

2.11. RNA extraction and DNase treatment

The overnight culture of *P. aeruginosa* grown in the presence and absence of the purified compounds were used for RNA extraction with DNase treatment using the NucleoSpin® RNA Midi kit (Macherey-Nagel, Germany). The extracted RNA was then subjected to reverse transcriptase PCR to ensure no contamination with DNA before subjecting to real-time PCR. The RNA concentrations were normalized to 50 ng/μl and stored at −80 °C until further use.

2.12. cDNA synthesis

cDNA synthesis was carried out using PrimeScript 1st strand cDNA synthesis kit (Takara, Japan). The protocol included incubation of the reaction mixture at 30 °C for 10 min followed by 1 h at 42 °C. Post incubation, the enzyme was inactivated at 70 °C for 15 min and cooled on ice. The cDNA obtained was stored at −20 °C until use.

2.13. Quantitative expression of virulence genes

Quantitative real-time PCR (qPCR) was performed using the SensiFAST SYBR No-ROX kit (Proteogen, India) according to the manufacturer's instruction. The genes *algD*, *lasB*, *piliA*, *flagA*, *PhzH* and *rhlI*, which are expressed under the control of the QS system, were selected. The primers were designed using Genscript primer designing tool and the sequence details are shown in Table S2. The optimization of the concentration was carried out for each set of primers. The real-time PCR master mix contained 10 μl of 2× SensiFAST SYBR mix, 1.6μlforward and reverse primers, 2 μl of template and water as required having a final volume of 20 μl. Melt curve analysis was performed at the end of 39 cycles to check for the presence of unique PCR reaction product. To detect the presence of contaminating DNA, a no-template control (NTC) was included. The ribosomal *rpsL* gene was used as a housekeeping reference gene [9]. Results were expressed as ratios of gene expression between the target gene and the reference gene. Analysis of relative gene expression was performed using the $2^{-\Delta\Delta ct}$ method [10]. Δct was calculated by the formula (average Ct of target − average Ct of reference) & similarly $2^{-\Delta\Delta ct}$ with the formula (Δct of target sample − Δct of reference sample). Data acquisition was performed by CFX96™ (Bio-Rad) manager at the end of each elongation step.

2.14. In silico analysis

2.14.1. Modelling platform

All computational analysis was carried out on Schrodinger 2018-3 suite device Maestro 11.7.012. This software package was programmed on DELL Inc.27 workstation machine running on Intel Core i7-7700 CPU@ 3.60 GHz ×8, a processor with 8GB RAM and 1000 GB hard disk with Linux–X6_64 as the operating system.

2.14.2. Molecular docking

Molecular docking was performed to know the interactions between the ligand and the receptor and to compare the affinities of all compounds to the target protein of *P. aeruginosa* QS signal receptor. For docking calculation, the protein (PDB ID: 3SZT) was downloaded from protein data bank and refined using the protein preparation wizard on the Schrodinger suite. All the ligand molecules were prepared by the ligprep tool. Docking score was calculated using Maestro 11.5 (Schrodinger) software. Glide Extra precision (XP) tool was used for the justification of suitable ligand molecule to the active site of the specific target. The binding affinity was assessed in terms of binding free energies (G-score, kcal/mol).

2.15. Statistical analysis

The data were subjected to *t*-test using SPSS, 16.0 software and was considered significant at $p \leq 0.05$.

3. Results

3.1. Compound separation

The effects of the compounds were first tested on the biomonitor strain of *C. violaceum*. The initial readout of anti-QS activity was based on the ability of the extract to inhibit the pigment formation of the biomonitor strain. Among the three fractions, the aqueous fraction of *C. verum* leaves (voucher number, NUCSER026) showed good anti-QS activity. Whereas, petroleum ether and chloroform fractions did not show any zone of pigment inhibition against *C. violaceum*. Based on this result, the subsequent characterization was carried out using the aqueous fraction of *C. verum* leaves.

3.2. Separation using column chromatography

A total of 450 elusions were obtained from the aqueous fraction. Based on the Rf value after thin-layer chromatography, fractions showing same reading were pooled and 10 major fractions were collected. These fractions were checked for their anti-QS activity against *C. violaceum*. Of the 10 fractions, the 9th fraction (tube no: 325-407 RF: 0.94) showed good anti-QS activity. The 9th fraction was subjected to further characterization and referred in the text as the purified compounds.

3.3. Lambda max

The active fraction exhibited UV max at 217 nm, 275 nm and 327 nm corresponding to $CH2{=}CH2$, C7H8, C6H14 respectively indicating the presence of 1,3-butadiene, toluene and hexane (Fig. 1A).

3.4. FT-IR spectral analysis

IR spectral data showed (C≡C–H: C—H stretch) at 3289.02 cm^{-1}, (C—H stretch) at 2921.01 cm^{-1}, (-C=C-) at 1640.08 cm^{-1} and (C—N stretch) at 1025 cm^{-1} indicating the presence of alkynes (terminal), alkanes, alkenes and aliphatic amines (Figs. 1B & S3).

3.5. GCMS spectral analysis

Spectra for GCMS showed the presence of 13 compounds thought to be responsible for anti-QS activity of *C. verum* leaf extract (Fig. 1C & Table 1).Complete report of GCMS analysis is provided in Supplementary material (S3).

3.6. Assessment of P. aeruginosa in the presence of purified compounds

Maximum inhibition of virulence factor production was found at a concentration of 0.1 mg/ml. Purified compounds were able to decrease the swarming motility (48.78%), pyocyanin production (81.10%), biofilm formation (86.56%) and elastase production (92.62%) in

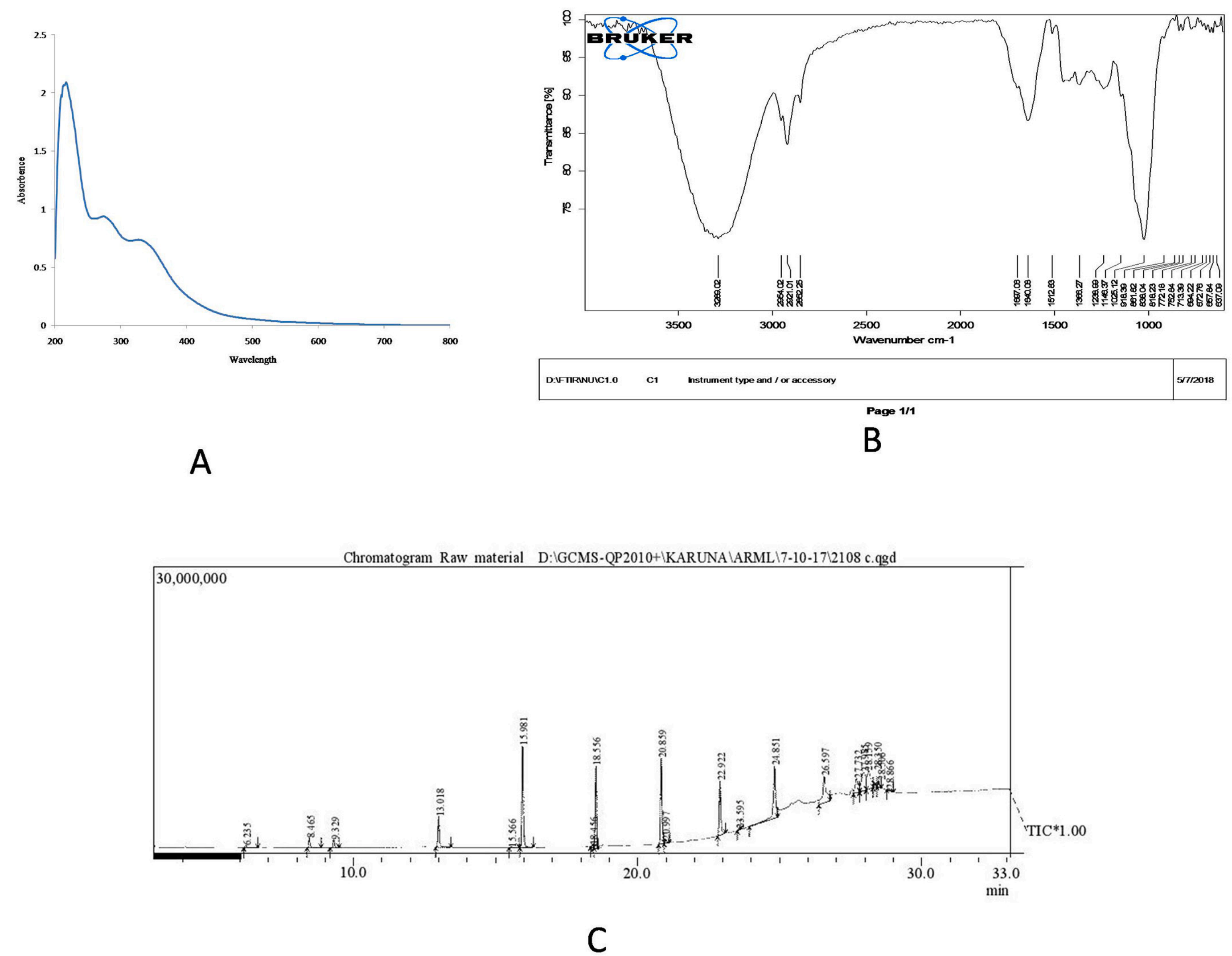

Fig. 1. Analysis of the purified compounds isolated from *C. verum* (A) spectra of Lambda max; (B) spectra of FT-IR spectroscopy; (C) GCMS spectra.

P. aeruginosa (Table 2).

3.7. Toxicity studies

To check for the toxicity of the compound, $LC_{50}$was calculated. The concentration at which all the embryos were dead (LC_{100}) and the concentration at which all the embryos survived (LC_0) were considered as the upper and the lower limit respectively for calculating $LC_{50.}$ As shown in Table S4 and Fig. 2A, the LC_{50} of the purified compounds was estimated to be 184 mg/l. Morphological deformities were evident at concentration of 180 mg/l onwards whereas concentrations above 260 mg/l were lethal. No morphological defects and embryonic mortalities were observed at concentration below 100 mg/l. Even at 100 mg/l, despite no death till the last day of observation (5th day), 2 embryos out of 20 in each replicate displayed incomplete yolk sac consumption. In case of 120 mg/l, 8 out of 20 embryos were lethal on the 5th day.

3.8. Determination of survivability

In the survivability assay, the number of embryos surviving on day 1 post exposure was recorded. Survivability of the embryos was found to decrease with the increased concentration of the purified compounds (Fig. 2B). 10% and 30% embryos were lethal on the 5th day at a concentration of 100 mg/l and 120 mg/l respectively. Similarly, at 140 mg/l, it was 17% and at 180 mg/l, it was 40% on the 3rd day. At 220 mg/l none of the embryos survived beyond 3 days.

3.9. Evaluation of cardiotoxicity

To check the cardiotoxicity of the compound, the heart rate and rhythmicity were analysed on day 2 post fertilization, a stage where the organogenesis is complete and the blood circulation is intact. The heart rate was calculated based on beats per minute. The values are expressed as mean of three independent observations for each concentration. No changes in the heart rate were found at concentrations up to 100 mg/l. As shown in Fig. 2C, at concentrations greater than 100 mg/l, the effects on cardiac functions were quite obvious. Drastic reduction in the heartbeat was found at 220 mg/l (60 beats/min) compared to the control. However, there was no change in the rhythmicity at the various concentrations tested.

3.10. Evaluation of hatchability

The ability of the embryos to break open the chorion and start swimming as free larvae is called hatching and the ability of the embryos to hatch on their own is a major developmental landmark. Here, upon exposure of the test compound, the hatching ability was found to be decreased with increased concentration of the purified compounds. As

Table 1
List of compounds reported in GCMS analysis.

Peak no	Name of the compound	Pubchem ID	Activities reported
1	Benzenemethanol, .alpha.,. alpha.-dimethyl-	57350271	Decreases the catalytic activity of cytochrome P4501A1 in *E. coli* [11]
2	Hydroperoxide, 1-methyl-1-phenylethyl	6629	Oxidizing agent [12]
3	Benzenamine, N,N-diethyl-4-methyl-	69177	–
4	Diethyleneglycoldimethacrylate	7355	–
5,6	2-Propenoic acid, 2-methyl-, 1,2-ethanediylbis(oxy-2,1-ethanediyl) ester	7979	Promotes the odontoclastic differentiation ability of human dental pulp cells [13]
7	Z,Z-3,13-Octadecedien-1-ol-	5364464	Antibacterial and antioxidant activity [14]
8,9,11	2-Propenoic acid, 2-methyl-, oxybis(2,1-ethanediyloxy-2,1-ethanediyl) ester	7980	–
10	Oleoyl chloride	5364783	Antimicrobial activity [15] Membrane – permeabilizing, endotoxin – neutralizing activity against *E. coli* and *Staphylococcus aureus* [16]
12	Heptaethylene glycol monododecyl ether	76495	–
13,14	9-Octadecenoic acid (*Z*)-, 2-butoxyethyl ester	6436064	Anti-inflammatory and anti-arthritic activity [17]
15,16,17,18	Oleic acid, 2-(1-octadecenyloxy) ethyl ester, (*E*)-	102446663	–
19	Cyclohexyl-15-crown-5	86572	Antimicrobial and antioxidant activity [18]
20	Stearic acid, 2-(1-octadecenyloxy)ethyl ester, (Z)	5378765	Antibacterial and antifungal activity [19]

Note: 9th pooled fraction from column chromatography was characterized and was found to contain 13 compounds which are together referred as purified compounds.

Table 2
Reduction in virulence factors of *P. aeruginosa* in the presence of purified compounds.

	Pyocyanin assay (μg/ml)	Swarming assay (mm)	Elastase assay (OD)	Biofilm assay (OD)
Control	5.335 μg/ml	8.33 ± 0.577	1.587 ± 0.016	1.25 ± 0.005
Exposed	1.007 μg/ml	4.266 ± 0.25	0.177 ± 0.006	0.168 ± 0.01

shown in Fig. 2D, the percentage of embryos that showed delayed hatching at concentration of 100, 120, 140 and 180 mg/l were 93, 90, 66 and 43% embryos respectively. In addition, 50% embryos at 140 mg/l and 13% embryos at m 180 mg/l had defective swimming behaviour. None of the embryos survived on the 3rd day at 220 mg/l.

3.11. Gene expression studies

The results of real time PCR revealed that the relative expression of virulence genes of *P. aeruginosa* were significantly lower ($p < 0.05$) than that of the control indicating the ability of the purified compounds to inhibit the expression of virulence genes associated with QS (Fig. 3). The highest reduction was seen in the expression of alginate and elastase B genes with a decrease of 15 and 14 folds respectively. The least reduction of 3 and 5 folds was observed in pili A and flagella A genes respectively.

3.12. In silico analysis

All compounds reported in GCMS analysis were docked in the groove of the binding sites present in 3SZT. The affinity of the compounds with the receptor is listed in Table S5 in terms of G-score. The compounds have binding free energy in the range of −7.104 to −1.686 kcal/mol. The active residues in3SZTwere Ser38, Gly 40, Ala 41, Arg 42, Pro 117, Val 78, Gly 81, Leu 82, Tyr 52, Phe 54, Tyr 66, Thr 72, Ser 129, Met 127, Ser 126, Ile 125.

Benzenamine, N,N-diethyl-4-methyl- (structure 3) has the best binding energy of −7.104 with the highest affinity among all other compounds. Cyclohexyl-15-crown-5, 2-propenoic acid, 2-methyl-oxybis 2,1-ethanediyloxy-2,1-ethanediyl ester and 9-octadecenoic acid (*Z*)-, 2-butoxyethyl ester fits into binding cleft of 3SZT receptor with G-score −6.42 kcal/mol, −6.367 kcal/mol and −5.876 kcal/mol respectively. Hepta ethylene glycol monododecyl ether (structure 9), oleic acid, 2-(1-octadecenyloxy) ethyl ester, (*E*)-(structure 11) and stearic acid, 2-(1-octadecenyloxy) ethyl ester, (Z) (structure 13) did not show any interaction.

The hydrogen bond interaction of structure 3 was formed with Ser 129 (Fig. 4A). Structure 12 formed hydrogen bond interaction with Ser 38, Trp 62, Tyr 58 (Fig. 4B). The hydrophobic interplay between the ligand and the receptor also represents good interaction. The docking orientation for structure 3 with the 3SZT receptor is represented in Fig. 4C.

4. Discussion

Due to continued and indiscriminate use of antibiotics, the emergence of drug resistance among bacteria has become one of the major threats worldwide. The past two decades have seen a significant surge in the antibiotic resistance. The rate at which new antimicrobial agents are being developed is way slower than that of emergence of antibiotic resistance. To overcome this growing problem of antibiotic resistance, development of alternative and effective strategies is of paramount importance. Attenuation of QS is one such strategy that could be targeted to combat drug resistance [20].

Herbal medicines have been used in the treatment of various ailments in traditional methods practiced in India such as Ayurveda, Unani, and Siddha. World Health Organization has reported plants to be the best source of antimicrobial compounds [21]. Like medicinal plants, dietary plants are also a rich source of phytochemicals and are generally considered to be safe with least toxicity. *C. verum* is one such medicinal as well as a dietary plant whose leaves are of culinary use in south western parts of India. To the best of our knowledge, this is the first study on isolation of anti-QS compounds against *P. aeruginosa* from *C. verum* leaves.

Based on the initial screening process that included 65 dietary plants, we selected *C. verum* ethanol leaf extract for isolation and characterization of anti-QS compounds. For the separation of anti-QS compounds from ethanol leaf extract of *C. verum* solvents were chosen based on their increasing polarity. As the study does not target a particular variety of phytochemical/s to be isolated, solvents with increasing polarity was used. Due to the diversity of plant bioactive compounds, the optimal solvent for the extraction depends on the plant material, and also the

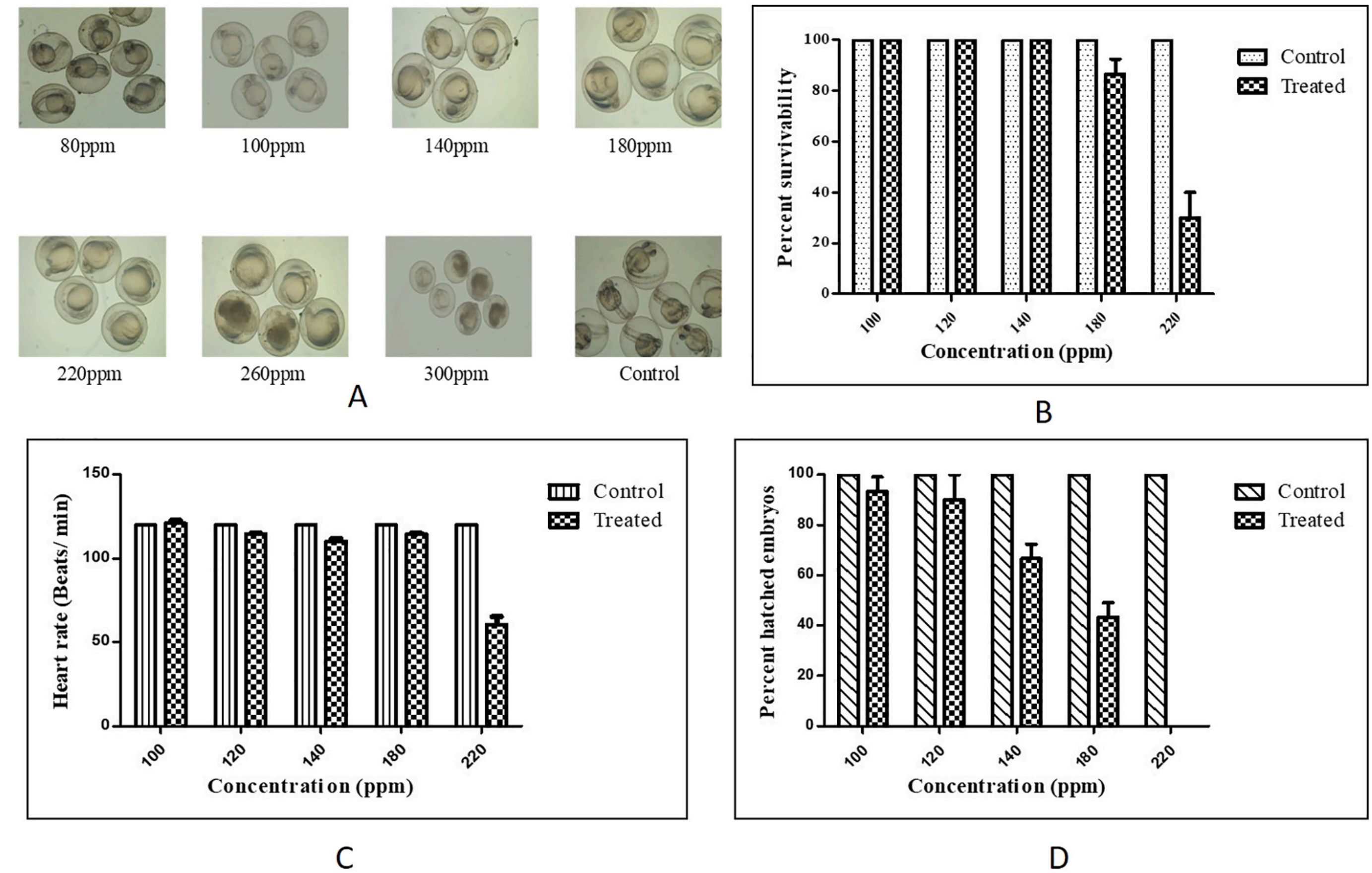

Fig. 2. Toxicity studies in zebra fish model at different concentrations of the compound (A) LC_{50} determination; (B) survivability of embryos on day 1; (C) heart rate; (D) hatching rate.
Note: Values were obtained in triplicates hence mean and SD has been calculated.

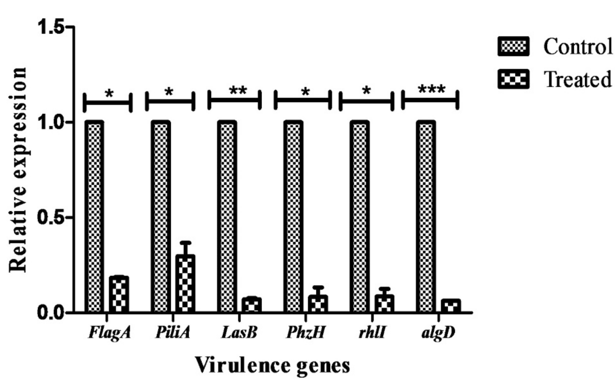

Fig. 3. Quantification of the virulence gene expression (*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$).

compounds to be isolated [22]. Petroleum ether is a nonpolar solvent useful in the extraction of alkaloids, terpenoids, coumarins, and fatty acids [23]. Terpenoids, flavonoids, fats and oils gets dissolved in chloroform [24]. Water is a polar solvent which is used to extract a wide range of polar compounds [25].

Only the aqueous fraction of *C. verum* leaf after solvent separation and column chromatography revealed the presence of anti-QS activity. It is well known that polar compounds are soluble with solvents of high polarity such as ethanol and water [26]. Therefore, the possibility of isolating anti-QS compounds is much higher in polar solvents.

Although, UV–visible spectroscopy is widely employed for the analysis of conjugated organic compounds its use is limited because of the inherent difficulties in assigning the absorption peaks to the constituent compounds [27].Thus, in the present study, UV-VIS findings were supplemented with other analytical techniques such as FT-IR spectroscopy and GCMS analysis, which enabled us in identifying and characterizing the compounds.

The UV absorption spectrum gives an idea of the carbon skeleton, functional groups, and the presence of conjugation in a molecule. As most of the compounds present in the purified fraction were long chain esters of fatty acids, their λ_{max} was found to be in the UV region. Strong absorption maxima found at 217 nm and 275 nm fell under the UVC region and another adsorption maximum at 327 nm in the UVA region. The first two λmax could be assigned to the n-π^* transitions of C—N and C═O groups respectively. The 327 nm absorption λmax owed to the n-π^* transitions of aromatic skeleton present in benenamine esters. The bathochromic shift was due to the extended conjugation of aromatic systems and other conjugated molecules.

The FTIR spectroscopy of the purified fraction indicated a broad band that appeared in the 3289 cm^{-1} regions. This could be attributed to the hydroxyl group stretching frequency of Z,Z, -3,13-octadecadien-1-ol. The absorption bands appearing in the range of 2852–2954 m^{-1} were related to olefinic C—H (HC═CH) as well as alkyl groups present in saturated/unsaturated fatty acid esters. The shift of the band towards the 2954 cm^{-1} region reflected the degree of unsaturation whereas the shift to the lower value of 2852 cm^{-1} indicated the level of saturation. The rule of three in explaining the FTIR bands due to ester groups state that the simple ester absorption bands appear at around 1700, 1200, and 1100 cm^{-1} due to C═O and C—O stretches. Owing to that in the recorded FTIR spectrum, three absorption bands appeared at 1697, 1238, and 1125 cm^{-1}. This data justifies the presence of alkyl ester

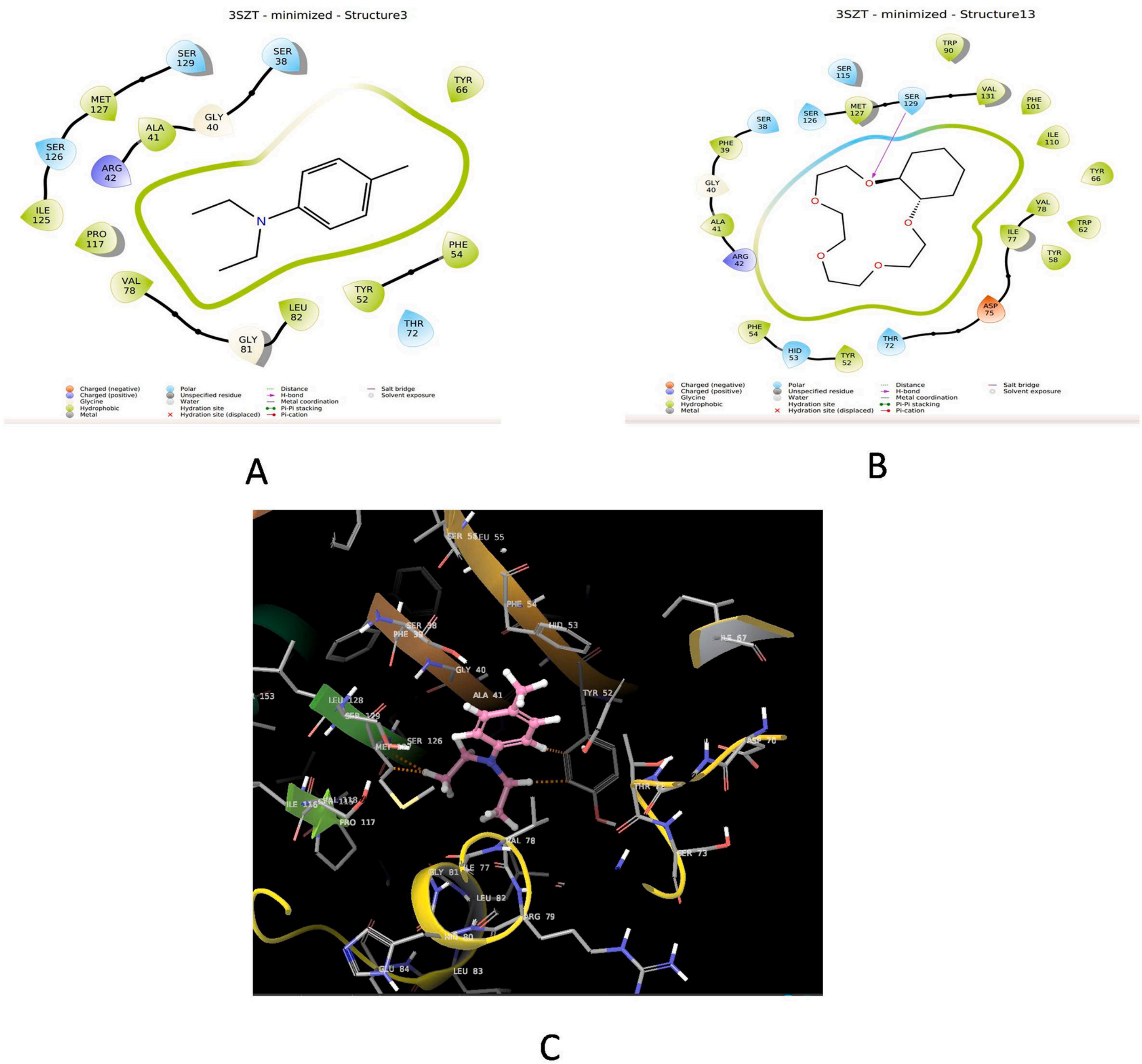

Fig. 4. Ligand interactions of (A) structure 3 with 3SZT; (B) structure 12 with 3SZT; (C) docking orientation of structure 3 with 3SZT.

groups in the active principle. Additionally, the absorption band seen at 1641 cm^{-1} was due to –C=C- stretch and out of plane bends for this group appeared at 1025 and 918 cm^{-1}. Another strong absorption band appeared at 1512 cm^{-1} confirmed C—N stretching might be due to the benzenamine derivatives. The remaining bands observed in fingerprint region were due to Ar—H bending and bending frequencies of other groups [28]. Thus, the data confirms the presence of the listed active principles in the purified fraction.

GCMS is an important analytical technique that enables the analysis of a mixture of organic compounds of low molecular weight. Based on the retention time and the mass spectra, 13 compounds were found to be present in the purified fraction of *C. verum* leaves. The identification of the compounds was carried out by using the National Institute Standard and Technology (NIST) database wherein, the mass spectrum of the unknown compound was compared with the spectrum of the known compounds from the NIST library [29].

The present study has applied *in silico* methodologies such as molecular docking for the analysis of protein-ligand interactions. Molecular docking studies explore the binding interaction between the compounds and the protein. The docking method used here was Glide XP, which estimates the protein-ligand interaction patterns and reveals the amino acid residues responsible for the binding. This method represents a single, coherent approach in which the sampling algorithms and the scoring function has been optimized simultaneously, and semi-quantitatively ranks the ability of candidate ligands to bind to a specified conformation of the protein receptor [30,31]. The interaction of ligands with the protein receptor (AHL) as revealed in the investigation clearly indicates the binding ability of the purified compounds to the autoinducer produced by *P. aeruginosa*.

To validate the results of *in silico* analysis, virulence assays and quantification of virulence genes was carried out. Phenotypic assays and gene expression studies authenticated the molecular docking data by indicating a decrease in the quantity of virulence factor produced along with the down regulation of specific virulence genes. Purified compounds isolated from *C. verum* in the current study were able to inhibit the virulence factor production in *P. aeruginosa* at a concentration of 0.1 mg/ml. Although, studies on *C. verum* purified compounds with anti-QS activity are limited, similar studies on other plant purified compounds acting as QS inhibitors against *P. aeruginosa* exist. Ellagic and ferulic acid isolated from the leaves of *Trapanatans*, showed anti-QS activity against *P. aeruginosa* by inhibiting the pyocyanin production (50%), biofilm formation (20%), elastase production (60%) at a concentration of 100 μg/ml [32].A study on sesquiterpene lactones recovered from an Argentine herb, *Centratherum punctatum* showed 50% inhibition of biofilm and elastase B production in *P. aeruginosa* at a concentration of 0.1 mg/ml [33]. A higher concentration of 800 μg/ml of phenolics isolated from the methanol extract of *Mangifera indica* leaves were used as anti-QS agents against *P. aeruginosa*. It has exhibited a reduction in the production of elastase (76%), total protease (56%), pyocyanin (89%), chitinase (55%), exopolysaccharide (58%), swarming motility (74%), and biofilm formation (72%) [34].The purified compounds of *C. verum* leaves in the present investigation were more effective at lower concentrations and were able to inhibit 80–90% of biofilm formation and elastase B in *P. aeruginosa*. Virulence assays performed in the presence of the purified compounds showed a marked reduction in the biofilm formation along with a decrease in the virulence factor production in *P. aeruginosa*.

Production of a majority of the virulence factors in *P. aeruginosa* is regulated by the QS system. To date, four QS systems namely, Las, Rhl, PQS, and IQS are identified in *P. aeruginosa*. Although, 4 QS systems have been reported in *P. aeruginosa*, Las and Rhl accounts for the majority of physiological processes and virulence phenotypes and constitute nearly 10% of the *P. aeruginosa* genome. Las and Rhl receptors are activated in *P. aeruginosa* by binding to N-oxododecanoyl-L-homoserine lactone and N-butyryl-L-homoserine lactone auto-inducers. LasR and RhlR are receptor proteins that in turn get activated to form complexes. This complex will further initiate the transcriptional expression [35]. As the genes responsible for the production of virulence factors is under the control of these QS systems, the disruption of QS system could therefore down regulate the expression of genes which in turn inhibit the production of virulence factors in *P. aeruginosa* [36].

To confirm the anti-QS activity of the purified compounds at the transcript level, qPCR was employed to quantify the expression of target virulence genes. The results of qPCR revealed a significant down regulation of the QS-regulatory gene *RhlI*, responsible for C4-HSL signal production, and other QS-regulated virulence genes such as *FlagA*, *PiliA*, *PhzH*, *LasB*, and *algD*. As the decrease in the production of biofilm and virulence factors are directly correlated with the down regulation in the expression of specific genes involved, it showed a direct evidence of the QS inhibitory activity of the purified compounds. The QS-regulatory gene, *RhlI* was down regulated by 12 folds in the treated conditions. Although there has been no studies so far that reported a reduction in the expression of the QS regulatory gene in *P. aeruginosa* upon treatment with compounds isolated from *C. verum*, the changes in gene expression level was comparable to findings obtained from studies on N-(2-pyrimidyl) butanamide and diallyldisulfide from garlic [37,38]. It can be conceived that the decrease in the expression of virulence genes, as observed in this study, can reduce the production of virulence factors thereby reducing the severity of the infection. The *algD* gene codes for alginate, which provides structural stability, water, and nutrient retention and protection of pseudomonal biofilms [39]. Since the expression of *algD* gene was down regulated in the presence of the purified compounds, it could be linked to the reduction in the production of mature biofilms, which in turn could facilitate antimicrobial substances to reach the bacterial cell wall. Similarly, the elastase B protein, which is encoded by *lasB* is responsible for degrading elastin containing lung tissue in the host [40]. The down-regulation of expression of *lasB* gene by the purified compounds could be an added advantage for the host defence mechanism to overcome pseudomonal infections. Pyocyanin, encoded by the *PhzH* gene, is an important virulence factor involved in the damage of multiple organ systems such as respiratory, urological, cardiovascular and central nervous system through the formation of reactive oxygen species [41]. *FlagA* gene, which codes for flagella, is required for adhesion to cells, swimming motility, and biofilm formation [42]. Pili encoded by *PiliA* gene, is involved in a form of surface movement termed as twitching motility. Twitching motility has been shown to be one of the factors required for the formation of biofilms on abiotic surfaces [43]. The expression of these genes which was significantly reduced in the presence of the purified compounds further validates the nature of action of the purified compounds as QS inhibitors. Although, the purified compounds did not completely inhibit the virulence factor production but their presence significantly lowered the virulence phenotypes as observed at both extracellular and transcriptional levels. The mechanism of action of the QS inhibitor could possibly be due to the repression of the QS-regulatory gene whose intracellular concentration was not sufficient enough to activate the genes involved in its motility, biofilm formation, elastase production, and pyocyanin synthesis. Thus, the study distinctly highlights the potential of the purified compounds from *C. verum* as QS inhibitors.

The *in vivo* toxicity of the purified compounds was expressed in terms of LC_{50}value, which is the dose that kills 50% of the tested population in an animal model [44].Concentration below 100 mg/L did not show any effect on zebrafish embryos which indicates that the characterized compound is non-toxic at lower concentrations. Although there are no reports of toxicity studies of plant purified compounds with anti-QS activity in the zebrafish model, studies on crude extracts of plants with antimicrobial properties and their toxicity assessment in zebrafish have been reported earlier. For instance, the LC_{50} value of ethanolic extract of *Punica granatum* peel in zebrafish was 196 mg/L [45], which is slightly higher than our observations. Similarly, another study by Alafiatayo et al. [46] reported that the teratogenic effect of *Curcuma longa* methanol extract was 62.5 mg/L in zebrafish. They also reported physical deformities such as kink tail, bend trunk, and the enlarged yolk sac at 96 and 120 hpf. In this study, the compounds from *C. verum* did not show any such effects at concentrations below 100 mg/L. Even at higher concentrations, the morphological and organ-specific effects were visible at later stages of development, suggesting the non-toxic nature of this compound. For any natural compound to be considered for potential applications, assessment of toxicity is mandatory in order to determine the associated adverse effects. Although phytochemicals or plant-derived compounds are presumed to be safe, thorough investigation of compounds, in terms of its safety, is often overlooked. Determination of LC_{50} value is a standard method to measure the lethal concentration of the compound. However, the effective concentration (EC_{50}) is generally referred to the concentration that produces the desired effect in at least 50% of the population, which may accompany some of the toxic effects. Thus a toxic concentration (TC_{50}) is usually a value between EC_{50} and LC_{50}. Our results indicated that concentrations below 100 mg/L is safe for the animals and concentrations between 100 and 184 mg/L resulted in deformities, suggesting a toxic dose. Since the focus in the study was to evaluate the toxicity of the compound, further studies would be necessary to determine the effective concentration of the compound, which could be less than 100 mg/L.

5. Conclusion

In the post-antibiotic era, the emergence of multidrug resistant pathogens has necessitated the search for alternative therapeutic strategies. The apprehending of the QS system could provide an effective and attractive anti-pathogenic therapy, to overcome the emergence of antibiotic resistance in bacteria. Our results demonstrate that the active fraction of *C. verum* leaves target the QS system of *P. aeruginosa* by acting as a potent QS antagonist. The compounds detected in the active fraction intervene in the QS system of *P. aeruginosa* and repress the genes responsible for the production of virulence factors. Furthermore, benzenamine, N,N-diethyl-4-methyl-, cyclohexyl-15-crown-5 and 2-

propenoic acid, 2-methyl-, oxybis (2,1-ethanediyloxy-2,1-ethanediyl) ester was identified as the probable compounds of the active fraction with the anti-QS activity based on the *in silico* analysis. Although, the active compounds described here qualify as a promising QS antagonist, further analysis would be necessary to validate the prospective pharmaceutical applications as anti-QS drugs.

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Balladka K. Sarojini: Formal analysis and Writing - review and editing.
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All authors read and approved the final version of the manuscript.

Declaration of competing interest

The authors declare that there is no known conflict of interest associated with this publication.

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